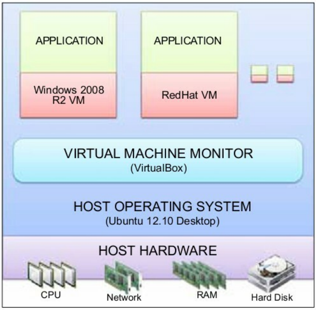

Figure 1: A virtual system

- **Shared folders:** With this feature, you can permanently or temporarily connect your host OS folders and use them as a network drive in your guest OS.
- **Snapshots:** A snapshot of the guest OS can be taken at any point. This helps you to revert the state of your guest OS in case something goes wrong with it.
- **State saving:** When you close the OS window, you are given the choice of sending a shut down signal, powering off, or saving the current state.
- **Seamless mode:** Like Parallels on Mac, this integrates the guest OS with your host OS' desktop.

The latest release of VirtualBox (v4.2.6) additionally comes with Windows Aero support and guest VM cloning facilities.

Installation and configuration on Ubuntu 12.10

VirtualBox can be installed on Windows, Solaris, Linux and the Mac iOS. In this section, we look at how to install and configure VirtualBox on Ubuntu 12.10 Desktop *(Quantal Quetzal)*. There are two ways of doing this. The first would be to manually download the VirtualBox DEB file from *https://www.virtualbox.org/wiki/Downloads* to your Ubuntu system and run it. However, this does not work all that well at first run; you may run into some complications like missing dependent packages or even issues with the kernel driver. All these can be solved, but take time and effort—so let's explore the easiest way. Simply launch a terminal window and run the commands given below.

> **Note:** In Linux, launch a terminal window using the key combination *Ctrl–Alt–T;* Mac users need to substitute *Ctrl* with the *Cmd* key.

The first step is to add a software repository key into the *sources.list.d/virtualbox.list* file:

```
wget -q http://download.virtualbox.org/virtualbox/debian/
oracle_vbox.asc -O- | sudo apt-key add -
```

Next, add the repository from where you are going to download VirtualBox:

```
sudo sh -c 'echo "deb http://download.virtualbox.org/
virtualbox/debian $(lsb_release -sc) contrib" >> /etc/apt/
sources.list'
```

Next, update your Ubuntu system with the necessary packages and updates, and then install VirtualBox:

```
sudo apt-get update
sudo apt-get install virtualbox-4.2
```

The installation will take some time to complete. Once done, you can start VirtualBox with the terminal command virtualbox, and you should have VirtualBox up and running.

```
sudo apt-get install linux-headers-3.5.0-17-generic
sudo /etc/init.d/vboxdrv setup
```

> **Note:** *'Failed, trying without DKMS'* is a very common error faced while installing VirtualBox on Ubuntu systems. To solve this, first download the correct Linux headers depending on the description of the error message, then run the *vboxdrv setup* command to recompile the kernel module and install it, and you should be able to launch VirtualBox successfully.

A Windows Server 2008 R2 VM

Once you have your VirtualBox set up, creating VMs is a fairly simple process. The following steps outline the VM creation process for Windows 2008 R2 Server. You'll need the Windows 2008 R2 Server ISO DVD image (you can download a 60-day trial from *http://technet.microsoft.com/en-us/evalcenter/dd459137.aspx*).

Launch VirtualBox and click the *New* icon to create a new VM. You will now see a *New Virtual Machine Wizard*, which walks you through the steps of selecting virtual resources for your Windows VM. The first dialogue box helps you select the type and version of the OS you are going to install as the guest OS. Provide a suitable name for the VM here as well. Once done, click *Next*, then select the amount of RAM you want to allocate to this VM.

> **Note:** Always set VM RAM to less than that of your host (physical) machine's RAM, since the host OS will require RAM for running itself and other applications.

The next step is to create the VM's hard disk. By default, it asks you to create a new hard disk from scratch using a *Create New Virtual Disk Wizard*. Select the correct *Type* of hard disk that you want for your VM, and provide a suitable size (in GB) for it.